

CoRE Stack — Technical Reference

Drought Algorithm, Input/Output Types, and Acronym Legend

computing/drought/generate_layers.py — github.com/core-stack-org/core-stack-backend

1. Drought Calculation — Mathematical Formulation

Spaces and index sets

R = FeatureCollection of micro-watershed polygons

$T = \{t_1, t_2, \dots, t_n\}$

monthly timesteps (monsoon season)

$Y = \{y_1, \dots, y_8\}$

agricultural years, 2017–2024

Satellite rasters (functions of space $\mathbf{x} \in \mathbb{R}^2$ and time)

$P : \mathbb{R}^2 \times T \rightarrow \mathbb{R}_+$

CHIRPS daily precipitation (mm/day)

ET, PET : $\mathbb{R}^2 \times T \rightarrow \mathbb{R}_+$

MODIS actual and potential evapotranspiration

$V, W : \mathbb{R}^2 \times T \rightarrow [-1, 1]$

MODIS NDVI and NDWI

$L : \mathbb{R}^2 \times Y \rightarrow \{1, \dots, 11\}$

LULC class label (CoRE Stack computed)

Spatial reduction

For a watershed polygon $r \in R$, define the area-weighted mean:

$$\mu(f, r, t) = \frac{1}{|r|} \int_r f(\mathbf{x}, t) \, dA$$

Five indices per watershed $r \in R$, period $t \in T$

(i) Standardised Precipitation Index (SPI) — meteorological drought

$$\text{SPI}(r, t) = \frac{\mu(P, r, t) - \mathbb{E}[\mu(P, r, \cdot)]}{\sigma[\mu(P, r, \cdot)]}$$

where expectation and standard deviation are taken over the historical record (CHIRPS, 1981–present).

(ii) Rainfall Deviation (D) — meteorological

$$D(r, t) = \frac{\mu(P, r, t) - \mathbb{E}[\mu(P, r, \cdot)]}{\mathbb{E}[\mu(P, r, \cdot)]} \in (-1, \infty)$$

(iii) Dry Spell (DS) — meteorological

$$\text{DS}(r, t) = \mathbb{1}[\forall w \in \{t, t+7, t+14, t+21\} : D(r, w) < -0.5]$$

A month is flagged as a dry spell if all four of its weeks show $> 50\%$ below-normal rainfall.

(iv) Vegetation Condition Index (VCI) — agricultural

$$\text{VCI}(r, t) = \frac{V(r, t) - \min_{t'} V(r, t')}{\max_{t'} V(r, t') - \min_{t'} V(r, t')} \in [0, 1]$$

Normalises current vegetation greenness against the full historical range (MODIS NDVI, 2000–present).

(v) Moisture Adequacy Index (MAI) — agricultural

$$\text{MAI}(r, t) = \frac{\mu(\text{ET}, r, t)}{\mu(\text{PET}, r, t)} \in [0, 1]$$

(vi) Percentage Area Sown (PAS) — agricultural

$$\text{PAS}(r, y) = \frac{|\text{kharif-cropped pixels in year } y|}{|\text{historically cropable pixels}|} \in [0, 1]$$

Derived from CoRE Stack LULC classes 8–11 (kharif crop types).

Final classification

A hardcoded decision table over $(\text{SPI}, D, \text{DS}, \text{VCI}, \text{MAI}, \text{PAS})$ produces an annual drought label:

$$\text{drlb} : R \times Y \longrightarrow \{0, 1, 2, 3\} \quad (0 = \text{none}, 1 = \text{mild}, 2 = \text{moderate}, 3 = \text{severe})$$

Drought frequency across the time series:

$$\text{freq}(r) = \frac{|\{y \in Y : \text{drlb}(r, y) > 0\}|}{|Y|} \in [0, 1]$$

Type signature

$$\text{drought} : (R, Y, P, \text{ET}, \text{PET}, V, W, L) \longrightarrow R \times (T \times Y \rightarrow \mathbb{R}^5) \times (Y \rightarrow \{0, \dots, 3\}) \times \mathbb{R}$$

2. Satellites and Data Sources

Acronym	Full Name	Description
MODIS	Moderate Resolution Imaging Spectroradiometer	NASA sensor on Terra/Aqua satellites; 250m–1km; daily coverage
CHIRPS	Climate Hazards Group InfraRed Precipitation with Station data	Rainfall: satellite + weather stations blended; 5.5km; daily since 1981
SRTM	Shuttle Radar Topography Mission	NASA elevation data from 2000 Space Shuttle mission; 30m global
FABDEM	Forest And Buildings removed Copernicus DEM	Improved DEM with trees and buildings removed; 30m
Sentinel-2	Sentinel-2	ESA satellite; 10m resolution; 5-day revisit
Landsat	Land Satellite	NASA/USGS satellite series; 30m; since 1972; 16-day revisit
JRC	Joint Research Centre	EU body; global surface water dataset tracking monthly extent since 1984
FLDAS	Famine Early Warning Systems Network Land Data Assimilation System	NASA modelled ET; fills spatial and temporal gaps in MODIS
GEE	Google Earth Engine	Google cloud platform for planetary-scale satellite imagery analysis
GCS	Google Cloud Storage	File store for raster/vector outputs exported from GEE

3. Spectral Indices

Acronym	Full Name	Formula / Meaning
NDVI	Normalised Difference Vegetation Index	$(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$ — vegetation greenness; 0 = bare, 1 = dense canopy
NDWI	Normalised Difference Water Index	$(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$ — surface water and soil moisture
NIR	Near Infrared	Wavelength band ≈ 800 nm; plants reflect strongly, water absorbs
ET	Evapotranspiration	Total water lost from land surface to atmosphere
PET	Potential Evapotranspiration	Water demand of the land if water supply were unlimited

4. Drought Indices

Acronym	Full Name	What it measures
SPI	Standardised Precipitation Index	Rainfall deficit vs 40-year historical mean/stddev; WMO standard drought index
VCI	Vegetation Condition Index	Current NDVI normalised to historical min/max; detects vegetation stress
MAI	Moisture Adequacy Index	ET/PET — how well the land's water demand is met (0–100%)
PAS	Percentage Area Sown	Fraction of historically-cropable land actually farmed this year

5. Geographic and Administrative Terms

Acronym	Full Name	What it is
MWS	Micro Watershed	Small drainage unit ($\approx$ 500–5000 ha); primary spatial unit in CoRE Stack
LULC	Land Use / Land Cover	Pixel classification: forest, agriculture, water, built-up, barren, etc.
DEM	Digital Elevation Model	Raster grid of elevation values (metres above sea level)
SOI	Survey of India	National mapping agency; source of administrative boundary data
AEZ	Agro Ecological Zone	Region with similar climate, soil, and farming conditions
CGWB	Central Ground Water Board	Indian government body; source of aquifer and groundwater data
IMD	India Meteorological Department	India's national weather service; issues official drought declarations
CLART	Catchment Land Area Ratio	Catchment area / stream length — proxy for groundwater recharge potential